

Samuel Peter

DEVOPS & CLOUD / CRITICAL INFRASTRUCTURE ENGINEER

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PROFESSIONAL PROFILE

DevOps & Cloud / Critical Infrastructure Engineer with 20+ years of progressive experience in national-grid operations, from Grid Control Operator to Principal Manager. Combines high-availability operational experience with hands-on work in CI/CD, infrastructure as code, cloud deployment, observability, automation, and software delivery. Technical work spans EMS/SCADA operations, AWS, Kubernetes, Terraform, Docker, GitHub Actions, Jenkins, TypeScript, Python, and API-driven systems. Operational initiatives have supported 99.9%+ availability and approximately 30% lower outage impact. Currently pursuing an MSc in DevOps & Cloud Computing.

TECHNICAL CAPABILITIES

DevOps & Delivery: CI/CD, GitHub Actions, Jenkins, build/test automation, release engineering, Git

Cloud & Infrastructure: AWS, Docker, ECR, Elastic Beanstalk, Kubernetes/EKS, Terraform, Infrastructure as Code

Reliability & Operations: EMS/SCADA, observability, monitoring, alerting, incident response, RCA, SLA operations, runbooks

Software & APIs: TypeScript, JavaScript/ES6+, Python, Java, Node.js, React 19, FastAPI, OpenAPI, REST APIs, Tailwind CSS, Vite

Testing & Quality: k6, Selenium, JUnit, unit/integration/performance testing, automated validation

Security & Grid Systems: RBAC, access control, operational security, NERC compliance, 330kV/50Hz grid operations

PROFESSIONAL EXPERIENCE

Principal Manager, Grid System Operations | Transmission Company of Nigeria (TCN) | Jul 2025 – Present

- Lead monitoring, incident response, operational risk, and change governance for national-grid services.
- Maintain 99.9%+ availability through observability, escalation, incident-management, and operational controls.
- Apply infrastructure-as-code and automation to standardize deployments and reduce manual intervention.

Senior Manager, Grid System Operations | Transmission Company of Nigeria (TCN) | Jul 2021 – Jun 2025

- Led critical outage response and restoration within SLA requirements across operational teams.
- Directed root-cause analysis and corrective actions associated with approximately 30% lower outage impact.
- Maintained EMS/SCADA observability and tuned alerts to reduce false positives.
- Trained 15+ operators and authored runbooks used for audits and emergency drills.

Manager, Grid System Operations | Transmission Company of Nigeria (TCN) | Jul 2018 – Jun 2021

- Directed real-time grid operations across high-voltage transmission assets.
- Improved monitoring and alerting workflows for earlier detection of abnormal conditions.
- Standardized operational procedures and documentation for incident handling and cross-team coordination.

Assistant Manager, Grid System Operator | Transmission Company of Nigeria (TCN) | Jan 2015 – Jun 2018

- Coordinated control-room activities, switching operations, incident handling, and restoration support.
- Contributed to monitoring-tool rollouts and translated control-room requirements into monitoring and response workflows.

Officer I, Grid Control Operator | Transmission Company of Nigeria (TCN) | Jul 2011 – Dec 2014

- Executed approved switching procedures and real-time system-health monitoring.
- Supported outage restoration, abnormal-condition escalation, and operational event tracking.
- Maintained audit logs and operational records for traceability and post-event review.

Officer II, Grid Control Operator | Transmission Company of Nigeria (TCN) | Jun 2005 – Jun 2011

- Performed real-time transmission-system monitoring in a 24/7 control-room environment.
- Executed approved operating and switching procedures and supported restoration during disturbances.

SELECTED PROJECTS

GridPulse 330 — DevOps Power Grid Telemetry & Contingency Simulation

TypeScript | React 19 | Tailwind CSS 4 | Vite 8 | SCADA/EMS | Real-Time Simulation | Vercel

- Designed and engineered an interactive SCADA/EMS-style console with simulated live telemetry for 330kV transmission and 50Hz grid-frequency scenarios.
- Modelled load-rejection dynamics, Ferranti overvoltage, threshold alarms, cascading-line protection, and operator dispatch responses.
- Implemented simulated dispatch playbooks for 150 MVAR shunt-reactor switching, -400 MW generation de-load, and corridor auto-reclosure.
- Added sequence-of-events/audit logging for transient review and RCA; versioned in Git and deployed through Vercel.

Live: <https://gridpulse-330app-szws.vercel.app/> | GitHub: <https://github.com/sampeter-akan/gridpulse-330app>

Movies API — Contract-First Microservices

TypeScript | Node.js | OpenAPI v3 | k6 | GitHub Actions

- Built REST microservices from an explicit OpenAPI v3 contract with automated validation and CI/CD.

- Used repeatable k6 performance tests and baselining to validate key endpoint response-time targets.

GitHub: <https://github.com/sampeter-akan/movies-api>

Credit Loan Calculator — Automated Build & Test Pipeline

Java | Jenkins | Maven | Selenium | JUnit | GitHub Actions | Dependabot

- Implemented automated build-and-test stages with Jenkins, Maven, Selenium, and JUnit.
- Added dependency-update automation and structured logging to support maintainable delivery.

GitHub: <https://github.com/sampeter-akan/credit-loan-calculator>

HomeStayGrid ML Recommender Feed

Python | FastAPI | Docker | AWS | ECR | Elastic Beanstalk

- Built and deployed a cloud-oriented ML recommender prototype using FastAPI, Docker, and AWS deployment workflows.
- Applied ML-generated recommendations to a maintenance-prioritization use case.

GitHub: https://github.com/sampeter-akan/homestaygrid_ml_recommender_feed

All Times Fitness Club — Front-End Project

HTML5 | CSS3 | Responsive Design

- Built a responsive front-end website using semantic HTML, CSS layout, and the original club branding.

EDUCATION

MSc DevOps & Cloud Computing — IU International University of Applied Sciences, Germany | Ongoing
GPA 80.5% | DevOps Engineering 96% | Advanced Research Methods 90% | Cloud Computing Project 84%

BSc (Hons) Computing & Information Technologies — University of Derby, UK | 2020
First Class Honours

BTEC Level 5 HND Diploma in Computing and Systems Development (QCF) — Pearson RDI, UK | 2014

LANGUAGES

English (C1) | Efik/Ibibio (Native)